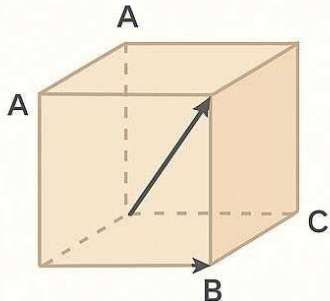
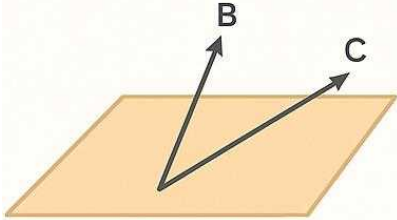


3.1 Triple Product Applications

A wide range of application work based on the triple products, which is primarily used to calculate the volume of a parallelepiped, making it essential in 3D geometry, architecture, and design, where accurate volume measurements are needed. While the vector triple product is used to simplify complex vector expressions, particularly in physics and mechanics, as shown in Table 5.

Table 5: **Triple Products Application**

NO	Field	Example	Shows
1	Volume of a Parallelepiped (3D shape)	Show the size of 3D shapes	
2	Checking Coplanarity	Validate flat surfaces or movements in a single plane.	
3	Mechanics and Dynamics	joint rotations and force directions	